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Examination paper for TDT4145 Data Modelling, Databases and Database Management Systems

Date: 24. February 2026

Time: 10:00-12:00 (10am-12am)

Course contact: Theodoros Chondrogiannis

Present at the exam location: YES (10am-12:00am)

Permitted examination support material: D: No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website "Explanation of grades and appeals".

SPECIFIC INFORMATION FOR YOUR COURSE

No paper drawings

This exam does not include hand drawings. If you receive hand drawing sheets, this is by mistake. **You will not be able to submit the sheets, and they will not be graded.**

Weighting:

The weighting is given in the heading of each problem.

Change the language of the question set:

If the question set is given in both English and Norwegian, you can change the language by going to the "hamburger menu" in the top hand-right corner and click "Language".

Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the "hamburger menu" in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspira.

1 Q1.1 Relational Operators (5%)

Which of the following is not a basic/fundamental operator in relational algebra?

Select one alternative:

- ☐ Projection.
- ☐ No other alternative is correct.
- ☐ Selection.
- ☐ Set union.
- ☐ Set intersection.



Maximum marks: 5

2 Q1.2 Keys (5%)

The minimal set of superkey is called:

Select one alternative:

- ☐ Discriminator.
- ☐ Candidate key.
- ☐ Minimal subkey.
- ☐ Primary key.
- ☐ No other alternative is correct.



Maximum marks: 5

3 Q1.3 Join Types (5%)

Select all statements from the following that are **true**.

Select all alternatives that apply:

- ☐ Every natural join operation can be replaced by a combination of the cross product and the selection operator. ✓
- ☐ The size of the result of a left outer join is always equal to the size of the left relation.
- ☐ The order in which inner joins are written does not affect the result of a query. ✓
- ☐ The result size of a full outer join is always smaller than that of the cross product.
- ☐ A full outer join always results in tuples that contain null values.

Maximum marks: 5

4 Q1.4 RA Result (5%)

You are given the following tables

employee

name	salary	manager
Alice	40000	John
Trevor	38000	John
Bob	35000	Trevor
Jim	35000	Trevor
Dorothy	36000	Trevor

person

f_name	salary	city
Siobhan	40000	Trondheim
Marine	35000	Zurich
Yiyu	35000	Athens

How many tuples does the following relational algebra expression return?

$\Pi_{manager}(\sigma_{salary < 40000}(employee \bowtie person))$

Write your answer here: (1).

Maximum marks: 5

5 Q1.5 SQL Query Result 1 (5%)

You are given the following table

employee

name	salary	manager
John	50000	NULL
Alice	40000	John
Trevor	38000	John
Bob	35000	Trevor
Jim	35000	Trevor
Dorothy	36000	Trevor
Samuel	36000	Alice

What is the result of the following SQL query?

```
SELECT count(*)  
FROM employee AS a, employee AS b  
WHERE a.salary < 40000
```

Write your answer here: (35).

Maximum marks: 5

6 Q1.6 SQL Query Result 2 (5%)

You are given the following table

employee

name	salary	manager
John	50000	NULL
Alice	40000	John
Trevor	38000	John
Bob	35000	Trevor
Jim	35000	Trevor
Dorothy	36000	Trevor
Samuel	36000	Alice

What is the result of the following SQL query?

```
SELECT b.salary  
FROM employee AS a JOIN employee AS b ON (a.manager = b.name)  
WHERE a.salary < 36000  
ORDER BY b.salary  
LIMIT 1
```

Write your answer here: (38000).

Maximum marks: 5

7 Q1.7 SQL and Relational Algebra (5%)

You are given the following schema:

bar(*bar_id*, *name*, *location*)

drink(*drink_id*, *name*, *price*)

serves(*bar_id*, *drink_id*)

where *serves.bar_id* references *bar.bar_id* and *serves.drink_id* references *drink.drink_id*. Which of the following relational algebra queries is equivalent to the following SQL query?

```
SELECT bar.name, drink.name
FROM bar, serves, drink
WHERE bar.bar_id = serves.bar_id
AND drink.drink_id = serves.drink_id
AND price > 20
```

Select one alternative:

- ☐ $\Pi_{bar.name, drink.name}(\sigma_{price > 20}(bar \bowtie serves \bowtie drink))$.
- ☐ $\Pi_{bar.name, drink.name}(bar \bowtie \sigma_{price > 20}(serves \bowtie_{serves.drink_id = drink.drink_id} drink))$.
- ☐ $\Pi_{bar.name, drink.name}(bar \bowtie_{bar.bar_id = serves.bar_id} serves \bowtie_{serves.drink_id = drink.drink_id} (\sigma_{price > 20} drink))$.
- ☐ No other alternative is correct.
- ☐ $\Pi_{bar.id, drink.id}(\sigma_{price > 20}(drink \times serves))$.

Maximum marks: 5

8 Q1.8 Terminology (5%)

Which term matches best the following description?

"A relation that is not part of the conceptual model but is made visible to a user as a 'virtual relation'"

Select one alternative:

- ☐ Decomposition.
- ☒ View. 
- ☐ Assertion.
- ☐ No other alternative is correct.
- ☐ Domain.

Maximum marks: 5

9 Q1.9 SQL Query Processing (5%)

When processing an SQL query (without any nested subqueries), which of the following clauses is executed last?

Select one alternative:

☐ WHERE.

☐ SELECT. 

☐ GROUP BY.

☐ FROM.

☐ HAVING.

Maximum marks: 5

10 Q1.10 Transactions (5%)

ACID is a set of properties that ensure database transactions are processed reliably even during

failures or concurrent access. These properties are **A**tomicity, **C**onsistency, (**I**solation, Indexing, Integrity, Irreversibility) , and **D**urability.

Maximum marks: 5

11 Q1.11 Foreign keys (5%)

Consider a table T that has been created using the following SQL code:

```
CREATE TABLE T (  
  tid varchar(20),  
  attr integer,  
  PRIMARY KEY (tid)  
)
```

You then attempt to create a second table by executing the following SQL code:

```
CREATE TABLE RT (  
  tid varchar(20),  
  attr varchar(20),  
  PRIMARY KEY (tid),  
  FOREIGN KEY (attr) REFERENCES T  
)
```

Which one of the following sentences is **true**?

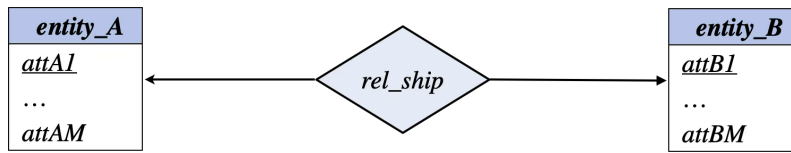
Select one alternative:

- ☐ The query fails because no attribute was given in the foreign key definition.
- ☐ The query is executed successfully; RT.attr references T.tid. 
- ☐ No other alternative is correct.
- ☐ The query is executed successfully; RT.attr of table RT references T.attr.
- ☐ The query fails because T.attr and RT.attr have different domains.

Maximum marks: 5

12 Q1.12 ER to Relations (5%)

Consider the following ER diagram:



What is the minimum number of relations (tables) needed to map this ER diagram to the relational model?

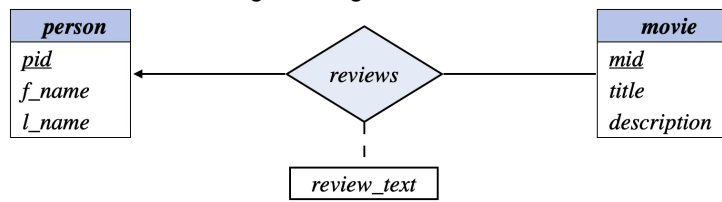
Select one alternative:

- ☐ No other alternative is correct.
- ☐ 2. ✓
- ☐ 3.
- ☐ 1.
- ☐ The ER diagram cannot be mapped to relations.

Maximum marks: 5

13 Q1.13 Constraints in ER (5%)

Consider the following ER diagram:



Based on this diagram, select all statements from the following that are **true**.

Select one or more alternatives:

- ☐ Each movie must have at least 1 review.
- ☐ Each movie can be reviewed by at most one person. ✓
- ☐ A person cannot submit two separate reviews for the same movie. ✓
- ☐ No two reviews can have the same review text.
- ☐ A person can review multiple movies; a movie can be reviewed by multiple people.

Maximum marks: 5

14 Q1.14 Normal Forms (5%)

Every relation that is in BCNF is also in:

Select one alternative:

- ☐ None of the other alternatives.
- ☐ First normal form.
- ☐ Third normal form.
- ☒ First, second, and third normal form.
- ☐ Second normal form.



Maximum marks: 5

15 Q2.1 - SQL DDL (10%)

You are given the following relational schema:

student(sid, first_name, last_name, email)

quiz(quiz_id, title, quiz_max_points)

question(quiz_id, quest_num, quest_text, quest_max_points)

answer(sid, quiz_id, quest_num, answer_text, points_gained)

Write the SQL code to create the table 'question' such that:

- quiz_id and quest_num are of integer type.
- quest_max_points is of numeric type with at most 2 digits and no decimals.
- quest_text is of varchar(100) type.
- The underlined attributes are primary keys.
- quiz_id is a foreign key referencing the attribute with the same name of table 'quiz'.
- The quest_text and the quest_max_points attributes cannot be NULL.
- The value of the max_points attribute should be in [0,20].

Write your answer in the box below. Changes are saved automatically.

1	<pre>CREATE TABLE question (quiz_id INTEGER, quest_num INTEGER, quest_text varchar(100) NOT NULL, quest_max_points NUMERIC(2,0) NOT NULL, PRIMARY KEY (quiz_id, quest_num), FOREIGN KEY (quiz_id) REFERENCES quiz, CHECK (quest_max_points >= 0 AND quest_max_points <= 20))</pre>
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Maximum marks: 10

16 Q2.2 SQL Query 1 (5%)

You are given the following relational schema:

student(*sid*, *first_name*, *last_name*, *email*)

quiz(*quiz_id*, *title*, *quiz_max_points*)

question(*quiz_id*, *quest_num*, *quest_text*, *quest_max_points*)

answer(*sid*, *quiz_id*, *quest_num*, *answer_text*, *points_gained*)

Write an SQL query that finds all quizzes in which the sum of the maximum points of their questions exceeds the *quiz_max_points* of that same quiz. Print the *quiz_id* of all such quizzes.

Write your answer in the box below. Changes are saved automatically.

1	<pre>SELECT quiz_id, quiz_max_points FROM quiz NATURAL JOIN question GROUP BY quiz_id, quiz_max_points HAVING sum(quest_max_points) > quiz_max_points</pre>
---	--

Maximum marks: 5

17 Q2.3 SQL Query 2 (5%)

You are given the following relational schema:

student(sid, first_name, last_name, email)

quiz(quiz_id, title, quiz_max_points)

question(quiz_id, quest_num, quest_text, quest_max_points)

answer(sid, quiz_id, quest_num, answer_text, points_gained)

Write an SQL query that computes the total number of points per quiz gained by the student with sid=12345, on quizzes which have title that contains the word 'databases'. Print the *quiz_id* and the total number of points gained on that quiz.

Write your answer in the box below. Changes are saved automatically.

1	<pre>SELECT quiz_id, sum(points_gained) FROM answer NATURAL JOIN quiz WHERE sid = 12345 AND title LIKE '%databases%' GROUP BY quiz_id</pre>
---	---

Maximum marks: 5

18 **Q2.4 SQL Query 3 (5%)**

You are given the following relational schema:

student(sid, first_name, last_name, email)

quiz(quiz_id, title, quiz_max_points)

question(quiz_id, quest_num, quest_text, quest_max_points)

answer(sid, quiz_id, quest_num, answer_text, points_gained)

Write an SQL query that prints for quiz with quiz_id=4325 all the *quest_text* of questions that have been answered by at least one student. Do not use aggregate functions in your answer.

Write your answer in the box below. Changes are saved automatically.

1	<pre> SELECT quest_text FROM question WHERE quiz_id = 4325 AND quest_num IN (SELECT quest_num FROM answer WHERE quiz_id = 4325) Alternative: SELECT quest_text FROM question NATURAL JOIN answer WHERE quiz_id = 4325 It was not required, a null check and an empty check are nice additions: SELECT quest_text FROM question NATURAL JOIN answer WHERE quiz_id = 4325 AND answer_text IS NOT NULL AND answer_text <> '' </pre>
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Maximum marks: 5

19 Q2.5 SQL Query 4 (5%)

You are given the following relational schema:

student(sid, first_name, last_name, email)

quiz(quiz_id, title, quiz_max_points)

question(quiz_id, quest_num, quest_text, quest_max_points)

answer(sid, quiz_id, quest_num, answer_text, points_gained)

You are also given the following SQL query:

```
SELECT first_name, last_name
```

```
FROM student
```

```
WHERE sid IN (
```

```
    SELECT sid
```

```
    FROM answer
```

```
    WHERE quiz_id = 4326
```

```
    AND question_num = 7
```

```
    AND points_gained > 10
```

```
)
```

Write an equivalent SQL query to the above without using nested queries.

Write your answer in the box below. Changes are saved automatically.

1	<pre>SELECT first_name, last_name FROM student NATURAL JOIN answer WHERE quiz_id = 4326 AND question_num = 7 AND points_gained > 10</pre>
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Maximum marks: 5

20 Q.0 Comments (0%)

This "task" is an opportunity to inform about circumstances, which are **absolutely necessary** to communicate to the censors, in order for them to make a correct judgement of your answers. This can for instance be assumptions that was completely unavoidable.

This is *not* for *general comments* about the exam, which can be stated on Ed or in email to the teachers.

Write your answer in the box below. Changes are saved automatically.

[illegible]

Maximum marks: 0